

BioDecisionBench: A Leak-Proof Typed-Decision Benchmark for Biological Sequences, and a Seed-Averaged Comparison of Decision Architectures

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Abstract

Through one interface, a model answers “Is this DNA sequence a promoter?” over candidates {yes, no}, “Which fold does this protein adopt?” over fold labels, and “How fit is this variant?” over five ordinal score levels; sequence, question and candidates are encoded as text and a single *shared scorer* returns one probability per candidate — no task-specific heads, no free-text generation. These interfaces are being adopted but never compared against task heads under matched compute. We release **BioDecisionBench**: 290 leak-proof biological sequence decision tasks spanning DNA, protein and RNA under four typed primitives (Noul/Choice/Score/ multi-Noul), with leak-proof admission (legacy blind tests preserved verbatim, 1,984 conflicting-label groups quarantined, zero within-task train/test overlap). We run the first seed-averaged comparison of three architectures: shared scorer, per-task heads, frozen small LM. Over three seeds, per-task heads match or beat the shared scorer (Noul AUROC 0.629 ± 0.050 vs. 0.478 ± 0.051 , disjoint SEMs; Choice and Score at parity) and the shared scorer is markedly less stable (run-std 0.087 vs. 0.030); its advantages survive only qualitatively (zero-shot at base level, weak cross-family transfer, caught by a 32-label head). Under a practical recipe the same benchmark yields near-ceiling baselines — single-task 0.671 TAPE fluorescence (99% of the published ceiling) and 0.845 promoter detection; one unified model 0.860 nuclear-protein positioning — showing joint-regime scores measure interference, not difficulty. Single-run differences of ± 0.05 –0.1 reverse conclusions; we document one reversal and propose ≥ 3 -seed averaging with run-std as the minimum bar.

1 Introduction

A single biological sequence model should ideally answer qualitatively different questions through one interface, and the interface can be as simple as text: present the sequence `GTATTCTCCCCTAACAAGAC...` with the question “*Is this DNA sequence a promoter?*” and the candidates {yes, no}, and a model returns $P(\text{yes})$; swap only the question and candidate set — {All Alpha, All Beta, Alpha and Beta, ...} for fold class, five value-anchored ordinal levels for mutation fitness, per-label yes/no for subcellular localization — and the same model, with no task-specific output heads and no free-text generation, answers all of them. *Typed-decision* interfaces implement exactly this design. They are being adopted in practice, by commercial systems (Almeida, 2026) and by open reimplementations (Stepanov et al., 2025; TianyuCodings, 2026; TheoLeeCJ, 2025), on the promise of one small model, dynamic candidate sets, and zero-adaptation-cost new tasks. Yet two basic questions have never been answered under controlled conditions. **Q1**: does a shared typed-decision scorer match or beat conventional per-task output heads at matched compute? **Q2**: does the “dynamic candidates / zero adaptation” selling point translate into a measurable performance or cost advantage?

Answering these questions runs into three obstacles. First, biological sequence evaluation is fragmented: GUE (Zhou et al., 2024), Genomic Benchmarks (Gresova et al., 2022), TAPE (Rao et al., 2019), ProteinGym (Notin et al., 2023) and friends each define their own task formats, splits and metrics, so decision architectures cannot be compared in one protocol. Second, leakage auditing is largely absent: several upstream “train” pools on public

hubs *contain the blind-test members of prior studies*, so a random re-split silently fabricates a new test set. Third — the methodological spine of this paper — *evaluation variance in this regime has never been quantified*. In our small-budget multi-task setting, two runs of the same configuration and seed differ by ± 0.05 – 0.1 on task-macro metrics, enough to reverse the winner of an architecture comparison (Section 5.1). Any single-run architectural conclusion is therefore unreliable, including our own early one.

Approach. We build a benchmark first and a measurement protocol second, then compare architectures. BioDecisionBench converts ten source families (290 tasks; DNA, protein, RNA, and double sequences) into one typed-decision record format with a leak-proof admission protocol (Section 3). On a 37-task matched slice we train the two comparable arms — a shared typed-decision scorer (#1) and per-task heads on the same encoder (#2) — with family-balanced interleaving, stabilized optimization, per-family dev early stopping, and three seeds; the frozen Qwen3-0.6B full-candidate likelihood (#3) serves as an off-the-shelf reference (Section 4). Section 5 reports the seed-averaged comparison and three operational tests of #1’s claimed value, plus training-dynamics and mechanism ablations.

Contributions.

- **BioDecisionBench** (Section 3): a 290-task, three-modality, four-primitive biological decision benchmark with a leak-proof admission protocol — exact cross-split sequence isolation, conflict-group quarantine (1,984 groups, not majority-voted), verbatim preservation of legacy blind tests, and label mappings validated on full pool $\cap$ test overlaps. Post-isolation within-task leakage is zero.
- **A seed-averaged evaluation protocol with a documented failure of single runs** (Sections 4.3 and 5.1): same-configuration reruns differ by up to $7\times$ on score Spearman (0.107 vs 0.015) and reverse the comparison outcome; we make ≥ 3 seeds, per-task seed means, SEM + run-std reporting, and dev early stopping the minimum bar for this class of comparison.
- **The first matched-compute three-way comparison of decision architectures on biology** (Sections 5.1 and 5.2): per-task heads $\geq$ shared scorer (Noul significantly better, Choice/Score at parity, and more stable across seeds); both trained arms beat the frozen candidate-likelihood baseline. The shared scorer’s advantage is *qualitative* — it can serve arbitrary new tasks with zero adaptation, but its zero-shot quality equals the untrained base, cross-family transfer is weak and classification-only, and ≈ 32 labels let a fresh head catch its zero-shot level.
- **Training-dynamics and mechanism diagnostics** (Sections 5.3 and 5.4): naive budget scaling destabilizes joint training below the untrained baseline; stabilization recovers but confirms a plateau; per-family checkpoint selection is a zero-inference-cost improvement (family-optimal steps span $26\times$); two reusable controlled ablations (noul candidate phrasing across three stable checkpoints; contrastive decoding for frozen VLM candidate likelihood).

Results preview. Seed-averaged, per-task heads reach Noul AUROC 0.629 against the shared scorer’s 0.478 (SEMs disjoint) — while a single earlier run had suggested the opposite ordering. The shared scorer’s zero-shot interface, its headline feature, is caught by a fresh 32-label linear head on 7 of 11 held-out tasks.

Figure 1 summarizes the benchmark and the headline comparison.

2 Related Work

Biological sequence benchmarks and foundation models. DNA benchmarks cover histone marks, promoters, splice sites and regulatory annotations (GUE/DNABERT-2, 28 tasks (Zhou et al., 2024); Genomic Benchmarks (Gresova et al., 2022); DeepSEA (Zhou and Troyanskaya, 2015); DeepSTARR (de Almeida et al., 2022)); protein benchmarks cover fluorescence, stability, homology and mutation-fitness assays (TAPE (Rao et al., 2019); ProteinGym’s 217 DMS assays (Notin et al., 2023); FLIP (Dallago et al., 2021)); localization and function benchmarks are multi-label (DeepLoc 2.0 (Thummuluri et al., 2022); DeepGOZero (Kulmanov and Hoehndorf, 2022)); RNA benchmarks cover RBP binding preference (RNAcompete (Ray et al., 2013)). Foundation models — ESM-style protein LMs (Lin et al., 2023), protein language tooling (Xu et al., 2023b,a), and multimodal DNA agents

0 train/test sequence leaks · 1,984 conflict groups quarantined
 290 leak-audited tasks = 10 source families x 4 primitives

ProteinGym [Prot]	—	—	217	—
GUE [DNA]	28	—	—	—
RNAcompete [RNA]	—	—	14	—
GenomicBenchmarks [DNA]	1	7	—	—
gene_lan [DNA+Prot]	—	8	—	—
local_snapshots [DNA+Prot]	5	2	—	—
dnagpt_pools [DNA]	3	—	—	—
TAPE [Prot]	—	—	2	—
DeepSTARR [DNA]	—	—	2	—
DeepLoc [Prot]	—	—	—	1
	choice	noul	score	multi-noul
	decision primitive (cell = number of tasks)			

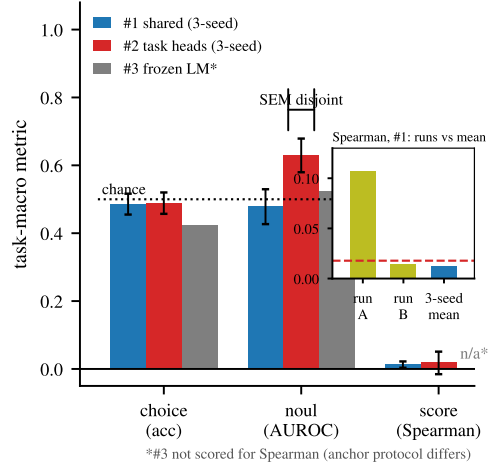


Figure 1: BioDecisionBench and its headline comparison. *Left:* 290 leak-audited tasks over 10 source families and 4 decision primitives; the admission protocol leaves zero within-task train/test sequence overlap (1,984 conflicting-label groups quarantined rather than majority-voted). *Right:* seed-averaged (3 seeds), matched-compute comparison per primitive (mean±SEM over tasks). Per-task heads (#2) beat the shared typed-decision scorer (#1) on noul (disjoint SEMs) and tie on choice/score; both trained arms exceed the frozen candidate-likelihood baseline (#3, not compute-matched). *Inset:* the methodological warning — two single runs of the same configuration give score Spearman 0.107 vs 0.015 for #1; the 3-seed mean is 0.013.

(de Almeida et al., 2025) — are trained and ranked per family. These resources are family-siloed: incompatible record formats, split protocols and metric conventions make cross-family, cross-interface measurement impossible, and most are single-sequence, single-task-type. BioDecisionBench is not another leaderboard substrate: it converts these sources into one typed-decision record format with leakage auditing as a first-class citizen (Section 3.3) and serves as the evaluation base for the architecture comparison.

Typed and candidate-scoring decision interfaces. The Jev line of systems (Almeida, 2026) exposes structured Noul/Choice/Score probabilistic judgements behind a service interface (weights not public). Open variants include GLiClass, a ModernBERT uni-encoder with dynamic label sets (Stepanov et al., 2025); NanoJev, a small shared-scorer reimplement (TianyuCodings, 2026); and SemIf, which reads option logits from a frozen LLM (TheoLeeCJ, 2025). Constrained decoding is the generative counterpart (Scholak et al., 2021). Each of these validates the interface’s feasibility; none compares *shared scorer vs. per-task heads* under matched compute, same data, and multiple seeds. We report the honest, counter-intuitive outcome of exactly that comparison.

Evaluation methodology and variance. Repeated runs and variance reporting are standard advice in ML evaluation (Bouthillier et al., 2021; Miller, 2024); LLM-evaluation work documents sensitivity to phrasing, prompt format and option order (Miller, 2024). In biology, prior work measured candidate-order sensitivity in the same decision setting: reordering candidates changes $\approx 9\%$ of predictions (Wang, 2026). This paper adds a concrete *conclusion-reversal* case for architecture comparison: two runs of one configuration give score Spearman 0.107 vs 0.015 for the shared scorer, flipping the winner; three-seed averaging overturns the single-run “shared $\geq$ heads” verdict (Section 5.1). To our knowledge this is the first fully documented instance of run variance reversing a decision-architecture conclusion, and we operationalize the remedy: ≥ 3 seeds, per-task seed means, SEM and run-std reported separately, and dev early stopping with ≥ 256 dev samples.

Probing and adaptation cost. Linear probing and few-shot head fitting are standard tools for characterizing representations. We operationalize the “zero-adaptation interface” claim as a *catch-up cost*: freeze the encoder, fit a new head for a held-out task with $N \in \{32, 128, 512\}$ labels, and report the smallest N at which the head reaches the shared scorer’s zero-shot level (Section 5.2). This turns “interface flexibility” from a qualitative claim into a comparable number.

Table 1: The four decision primitives, their extra fields, supervision and metrics.

Primitive	Extra fields	Supervision / evaluation
choice	candidates (mutually exclusive labels), answer	CE; accuracy, macro-F1, NLL, candidate-validity rate
noul	candidates = {no, yes}, answer	binary CE; AUROC, MCC, accuracy, calibration
score	score_levels (5), anchors (5 values from 4 interior train quantile cuts), gold_value, gold_level	soft-target CE + expected-value MSE; Spearman (primary), train-std-normalized RMSE
multi_noul	candidates (all labels), per-label 0/1 vector, answer = positive set	masked binary; micro/macro-F1, per-label calibration

Table 2: BioDecisionBench composition. Rows are the 10 source families (canonical task-id prefix map used by the training scheduler; `lg_/protein_homology` share the `local_snapshots` family). Task counts are per evaluation unit; ProteinGym’s 217 assays share one question type. Admission: 0 train/test sequence leaks after isolation; 1,984 conflicting-label groups quarantined (not majority-voted); 925,985 cross-task duplicate sequences kept isolated per task.

Family	Mod.	choice	noul	score	multi	Tasks	Train	Dev	Test
ProteinGym	P	–	–	217	–	217	415,201	140,300	140,810
GUE	D	28	–	–	–	28	731,069	82,350	82,125
RNAcompete	R	–	–	14	–	14	275,449	33,610	33,476
GenomicBenchmarks	D	1	7	–	–	8	516,071	57,251	162,541
gene_lan	D+P	–	8	–	–	8	104,130	12,897	12,973
local_snapshots	D+P	5	2	–	–	7	69,469	6,894	7,048
dnagpt_pools	D	3	–	–	–	3	111,269	12,527	13,892
TAPE	P	–	–	2	–	2	74,955	7,874	40,042
DeepSTARR	D	–	–	2	–	2	804,576	81,140	82,372
DeepLoc	P	–	–	–	1	1	16,878	5,462	5,963
Total	D42+P226+R14+8D	37	17	235	1	290	3,119,067	440,305	581,242

3 BioDecisionBench: Design and Construction

3.1 Unified interface and record format

Every example is one JSON line carrying `task_id`, `primitive`, `modality`, `sequences` (each `{role, modality, sequence}`), `question`, `split`, `group`, and `provenance`, plus interface fields determined by the decision primitive $p \in \{\text{noul}, \text{choice}, \text{score}, \text{multi_noul}\}$ (Table 1). Sequences keep role boundaries for double-sequence tasks (`query_1/query_2`: DNA+protein, DNA+DNA, protein+protein) rather than being concatenated. `provenance` records source repository, revision/commit and license; download receipts include per-file SHA-256.

3.2 Sources, scale, and modalities

Table 2 lists the composition. ProteinGym contributes 217 mutation-fitness assays using the official contiguous-5 per-mutant folds (fold 0 = test, fold 1 = dev, folds 2–4 = train; multi-mutant rows excluded); GUE 28 DNA choice tasks; RNAcompete 14 RNA score tasks (14-RBP subset, probes subsampled to 25k with a fixed seed); Genomic Benchmarks 8; `gene_lan_transfer` 8 double-sequence noul tasks with endpoint union-find component grouping; `dnagpt` DNA pools 3 (+1 promoter); TAPE 2; DeepSTARR 2 (dual outputs split into two tasks); DeepLoc 2.0 as one multi-noul task; and 7 local snapshots from earlier public releases (Wang, 2024, 2026). Interface distribution: choice 37 / noul 17 / score 235 / multi_noul 1; modality: DNA 42 / protein 226 / RNA 14 / double-sequence 8; rows: 3,119,067 train / 440,305 dev / 581,242 test.

Counting convention. The 290 figure counts evaluation units, not question types: ProteinGym’s 217 assays share one question type. We report both numbers wherever the benchmark scale is quoted and never claim “290 independent questions”.

3.3 Leak-proof admission protocol

1. **Native splits first.** Sources with official train/dev(/test) splits are used verbatim (GUE, Genomic Benchmarks, TAPE, DeepSTARR, ProteinGym folds, local snapshots).

2. **Legacy blind tests isolated from upstream pools.** The dnagpt pools (splice, TF, core) ship as unsplit “train” data *containing the blind-test members of prior studies* (Wang, 2026). Handling: test = legacy blind test, kept verbatim; train/dev = the pool with *all* legacy-test members removed, bucketed by sequence hash; rows whose pool integer label conflicts with the blind-test text label (8 in core) are quarantined as a group. Label mappings are not guessed but validated on full pool $\cap$ test overlaps (TF 3437/3437; core 5910/5918; splice agrees 4545/4545 under the legacy convention 0 $\rightarrow$ Non-Splice/1 $\rightarrow$ Acceptor/2 $\rightarrow$ Donor, while the literature’s newer builder uses a different convention — we adopt the data-consistent one and flag the acceptor/donor semantics for independent verification).
3. **Exact cross-split sequence isolation.** A sequence appearing in several splits is assigned wholly to the highest-priority split (test > dev > train); lower-split copies are dropped. Same-sequence conflicting labels quarantine the whole group (1,984 groups; virus_covid contributes 1,754) rather than majority-voting. We do *not* merge reverse complements: RC equivalence flips labels for strand-sensitive tasks (splice donor/acceptor, promoter direction); label-aware RC grouping is left as future work.
4. **Double-sequence tasks grouped by endpoints.** gene_lan pairs are split by union-find connected components over endpoints, so no endpoint appears in both train and test. A construction-diagnosable subset (the two “random” pairing tasks are separable by a frame-0 codon-table translation) is declared in the task notes and the paper, and is not claimed as an independent blind test.
5. **Score anchor discipline.** The 5-level anchors are derived from training data only (4 interior quantile cuts $\rightarrow$ 5 representative values); test values never feed back. Raw RMSE is never averaged across assays (native scales differ by orders of magnitude); Spearman is the primary metric.
6. **Admission checker.** Per task we verify split overlap, label conflicts, answer $\in$ candidates, and length distributions; globally we count cross-task duplicate sequences (925,985 sequence keys occur in >1 task — not within-task leakage under a multi-task model, but they must be escalated to global isolation under leave-one-task-out). Final admission result: **0 within-task leaks, 0 flagged tasks.**

3.4 Scope boundaries

Under the current encoder path (single sequence, DNA/protein, 512-token budget), 241/290 tasks are consumable by the compared architectures; the remaining 49 (14 RNA, 10 double-sequence, 1 multi-noul, 24 over-budget) are in the benchmark, covered by the corresponding modality/interface evaluation paths or reported with exclusion rates — never silently truncated. The benchmark’s value is the protocol and the data; it is not bound to any single evaluated architecture.

4 Evaluation Framework

4.1 The three compared architectures

#1 implements the typed-decision ideal: candidates are dynamic text, the scorer weights are shared across all tasks, and a new task needs no parameters. The encoder initialization is a reported protocol variable: the base tokenizer-expanded model (**no_cpt**, the clean-transfer regime of the matched comparison) or the biology continued-pretrained checkpoint (**cpt**, used by the practical-recipe leaderboard in Section 5.5). #2 is the conventional alternative on the same encoder. #3 answers “what does an off-the-shelf small LM get, for free?” and is reported for reference only.

4.2 Training protocol

Joint training interleaves families: each family contributes an equal number of batches per epoch (rotating within family), so ProteinGym’s 217 assays cannot flood the diverse small families. AdamW with encoder lr 2×10^{-5} , head/scorer lr 1×10^{-4} ; warmup fraction 0.15 and gradient clipping 0.5 — values set by the instability diagnosis of Section 5.3, not cherry-picked post hoc; cosine decay to 10%; batch 64 with gradient accumulation; 512-token full inputs (over-budget examples are excluded and the exclusion rate reported). Checkpoint selection: **per-family dev early stopping** (≥ 256 dev samples per family; each family keeps its own best-dev checkpoint from the same trajectory — zero inference-time cost), motivated by the $26\times$ spread of family-optimal steps (Section 5.3).

Table 3: Compared decision architectures. #1 and #2 are matched-compute (same data, updates, optimizer, seeds; the only variable is the decision mechanism). #3 is *not* matched (frozen, different backbone, 120-character sequence truncation, different score protocol) and is labeled as such wherever it appears.

	#1 shared typed-decision scorer	#2 encoder + per-task heads	#3 frozen candidate likelihood
backbone	Laya ModernBERT-large (Warner et al., 2025; NandhaKishorM, 2026), 423M, biology-expanded BPE (52,412)	same as #1	Qwen3-0.6B (Yang et al., 2025), 596M causal LM
decision	2-layer typed transformer + shared scalar scorer over candidate markers; choice pools over a candidate-set attention; score emits a 5-level distribution + expectation	one Linear head per task: choice/noul = fixed-class softmax; score = scalar regression (MSE)	length-normalized logprob per candidate, arg max
task parameters	none (candidates are input text)	one $ \mathcal{Y}_t $ -way head per task	none (frozen)
new task	inference with zero adaptation	requires a new head + labeled training	inference with zero adaptation
training	matched budget	matched budget	none

Table 4: Matched-compute three-way comparison, seed-averaged over 3 seeds. Mean $\pm$ SEM across tasks after per-task seed averaging; run-std is the across-seed standard deviation of the task-macro aggregate. #3 (frozen Qwen3-0.6B candidate likelihood) is *not* compute-matched and is reported for reference.

Primitive	Metric	n	#1 shared	#2 task heads	#3 frozen LM	run-std #1	run-std #2
choice	accuracy	21	0.486 $\pm$ 0.031	0.489 $\pm$ 0.031	0.424	0.040	0.030
noul	AUROC	7	0.478 $\pm$ 0.051	0.629 $\pm$ 0.050	0.522	0.087	0.030
score	Spearman	9	0.013 $\pm$ 0.009	0.018 $\pm$ 0.033	—	0.064	0.054
Per-task seed-mean wins (#1 / tie / #2)							4 / 23 / 10

4.3 Seed-averaging protocol

Motivating case (Section 5.1): two runs of the same configuration and seed give score Spearman 0.107 vs 0.015 for #1, reversing the comparison. Protocol: ≥ 3 seeds (20261001/02/03); average per task across seeds *first*, then aggregate across tasks; report mean $\pm$ SEM (across tasks) and run-std (across seeds) separately; early-stop dev sets ≥ 256 to shrink selection noise; any single-run number is labeled as such. GPU nondeterminism (identical-seed trajectories diverge after ~ 137 steps) is the physical source and cannot be removed by fixing seeds — only absorbed by averaging.

4.4 Metric conventions

choice: accuracy (primary) + macro-F1. noul: AUROC (primary — ranking-based, immune to threshold and calibration) + MCC + accuracy. score: Spearman (primary, scale-free) + RMSE normalized by train std; **raw RMSE is never averaged across assays**. multi_noul: per-label sigmoid protocol (prepared in the benchmark; not consumed by the encoder path in this paper) (calibration protocol follows Guo et al., 2017).

5 Results

5.1 Main comparison: seed-averaged three-way

The matched slice has 37 tasks (21 choice, 7 noul, 9 score; ProteinGym and GUE capped at 12 each), 2,520 updates, 3 seeds, per-family early stopping. Table 4 and the right panel of Figure 1 give the result.

Reading. (i) #2 $\geq$ #1: noul is significantly better (AUROC 0.629 $\pm$ 0.050 vs 0.478 $\pm$ 0.051, disjoint SEMs), choice and score are at parity (0.489 vs 0.486; 0.018 vs 0.013), and #2 is more stable across seeds (noul run-std

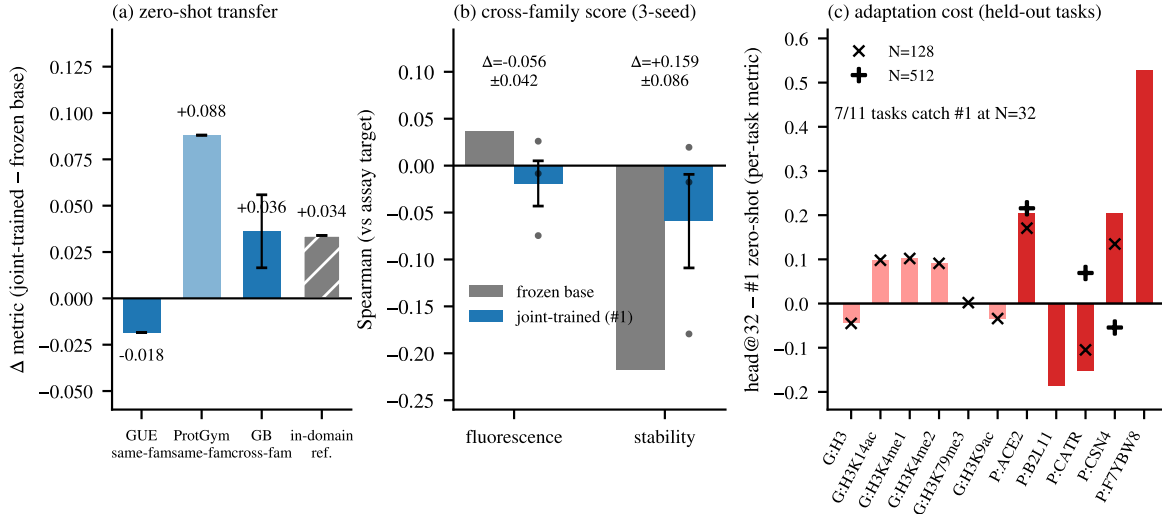


Figure 2: Three operational tests of the shared scorer’s claimed advantages. (a) Zero-shot transfer to same-family held-out tasks is a wash (GUE -0.018 , ProtGym $+0.088$); cross-family transfer on GenomicBenchmarks is weakly positive ($+0.036$, SEM over 3 seeds), comparable to the in-domain reference ($+0.034$; full2 configuration, not strictly matched). (b) Cross-family *score* transfer (TAPE, 3 seeds) does not hold: fluorescence degrades ($\Delta = -0.056 \pm 0.042$); stability’s apparent gain ($+0.159 \pm 0.086$) is regression of a pathological frozen baseline (-0.218) toward zero — trained absolute Spearman stays negative (-0.059). (c) Adaptation cost: a fresh #2 linear head catches #1’s zero-shot on 7/11 held-out tasks with only $N=32$ labels ($\times$: $N=128$, $+$: $N=512$).

0.030 vs 0.087; also 0.030 vs 0.040 on choice, 0.054 vs 0.064 on score). Per-task seed-mean wins: #1 4 / ties 23 / #2 10. (ii) Both trained arms beat the frozen #3 on choice (0.49/0.49 vs 0.42) and on noul for #2 (0.629 vs 0.522) — but #1 *loses* to the frozen small LM on noul (0.478 vs 0.522): a trained shared scorer ranks binary candidates worse than an untrained model reading their likelihoods. (iii) Absolute levels are modest (choice ≈ 0.49 , near majority-class baselines): this is the joint-37, 2,520-update, 512-window stress regime, calibrated against specialized-training anchors in Section 6.

Robustness to encoder initialization. The entire protocol was repeated on the biology-CPT encoder init (Section 4.1): the ordering is invariant — per-task heads reach noul AUROC 0.651 ± 0.042 against the shared scorer’s 0.519 ± 0.020 , choice 0.500 vs 0.485, score 0.056 vs -0.001 (per-task wins 3/22/12; Table 9). CPT lifts both arms on noul ($+0.041/+0.022$) without closing the gap, so the architecture conclusion does not depend on the encoder init.

The variance case. A single earlier run (before the seed protocol existed) reported #1 score Spearman 0.107 vs #2 -0.007 and a “#1 $\geq$ #2” overall verdict. A rerun of the identical configuration and seed fell to 0.015 — on *score* tasks, which do not touch the only code difference between the runs (noul candidate phrasing), so the gap is pure run variance; per-family early-stop steps drifted (GUE 200 $\rightarrow$ 5000, GB 1200 $\rightarrow$ 200). With 3-seed averaging, both arms converge to ≈ 0.01 – 0.02 and the “#1 clearly wins score” claim evaporates. This is the concrete failure mode Section 4.3 protocolizes against.

5.2 Three operational tests of #1’s claimed value

(a) Same-family zero-shot. Held-out tasks from families already in training (16 GUE + 20 ProteinGym tasks unseen by the 37-task joint model): #1-trained $\approx$ #1-frozen base (GUE accuracy 0.496 vs 0.515, $\Delta = -0.018$; per-task 11 wins / 9 ties / 13 losses — a wash). Joint training on 37 tasks did not learn a strongly transferable decision function. #2 cannot participate at all: it has no head for unseen label sets — the qualitative difference stands, the quantitative one does not.

(b) Cross-family zero-shot. Holding out the entire GenomicBenchmarks family (training on the other six families) and evaluating its 8 tasks gives mean $\Delta = +0.036 \pm 0.034$ over 3 seeds (per-seed $+0.038/-0.006/+0.077$)

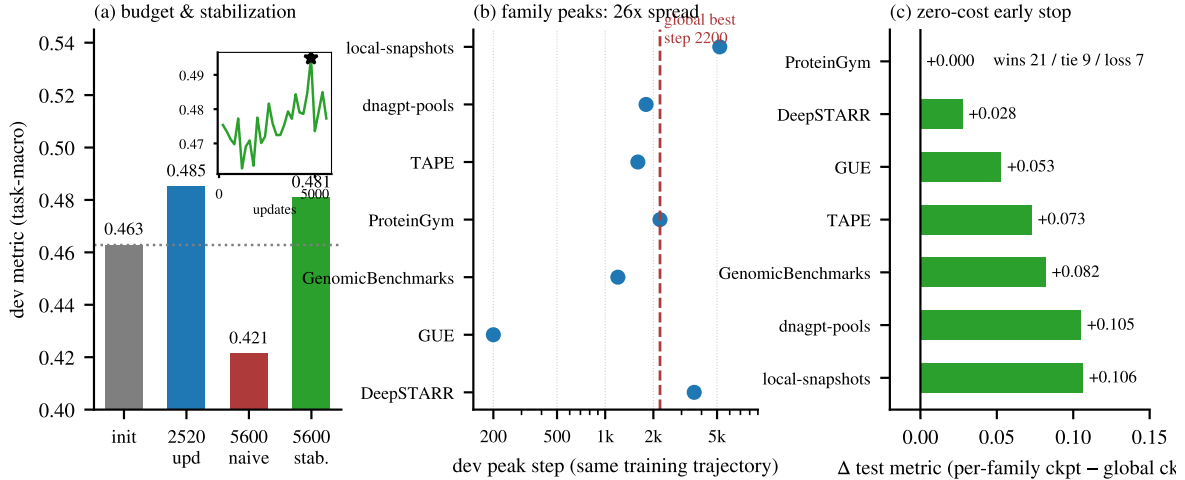


Figure 3: Training dynamics under joint multi-task fitting. (a) Naive budget scaling destabilizes: 5,600 updates score *below* init (0.421) while 2,520 reaches 0.485; stabilization (warmup 0.15, grad clip 0.5, dev early stop) recovers 0.481 at 5,600 — a plateau, not overfitting (inset: stabilized dev curve; star = selected checkpoint). (b) Family-optimal dev peaks on the *same* trajectory span steps 200–5,200 (26 $\times$), while the global best sits at 2,200. (c) Per-family early stopping (zero inference-time cost) dominates at family level (all $\Delta \geq 0$, mean +0.063); per task: 21 win / 9 tie / 7 loss vs the global checkpoint.

— the same magnitude as directly training on GB in-domain (+0.034), hinting the gain is a generic decision improvement rather than family-specific fitting, but the std matches the mean and per-task effects range -0.19 to $+0.31$. For the *score* interface, holding out TAPE (while training on the DeepSTARR and ProteinGym score families) over 3 seeds shows no clean transfer: fluorescence degrades ($\Delta = -0.056 \pm 0.042$, negative in all three seeds), and stability’s apparent $+0.159 \pm 0.086$ is entirely regression of a pathological frozen baseline (-0.218) toward zero — the trained absolute Spearman remains negative (-0.059). Cross-family transfer exists, weakly, for classification only.

(c) Adaptation cost. Freezing #2’s trained encoder and fitting a fresh linear head for each of 11 held-out tasks (6 GUE + 5 ProteinGym) with N labels: the new head catches or beats #1’s zero-shot on **7/11 tasks at $N = 32$** (Table 6); #1 stays ahead on 3/11 even when #2 is given $N = 512$; one task needs $N = 512$. Family means: GUE 0.476 (zero-shot) vs 0.511 (head@32); ProteinGym -0.024 vs 0.095. Thirty-two labels is a trivial cost — the “zero-adaptation” selling point does not convert into a meaningful performance-or-cost advantage.

(d) Noul phrasing on stable checkpoints. Candidate text is an inference-time degree of freedom that #1 has and #2 cannot have. A controlled ablation (same checkpoint, only the rendered candidate text varies; six phrasings) on the earlier single checkpoint had suggested raw “no/yes” beats the redundant “false: no / true: yes” by $+0.066$ AUROC. Re-running on three retrained seed-averaged stable checkpoints (Figure 4a) *reverses* the direction: the redundant prefix form wins on all three seeds (0.553 ± 0.023 vs the trained phrasing’s 0.508 ± 0.024), and **no phrasing reaches #2** (best 0.553 vs 0.629). Two conclusions: the noul deficit of #1 is not a phrasing artifact (the main result is reinforced), and even inference-level “clean controlled” evidence is checkpoint-dependent — ablations need multi-checkpoint confirmation too.

5.3 Training dynamics

Budget scaling is negative. On the same 43-task set, doubling updates 2,520 $\rightarrow$ 5,600 (and per-task data 768 $\rightarrow$ 2048) drops classification accuracy 0.485 $\rightarrow$ 0.421 — *below* the untrained init (0.463) — and TAPE Spearman $0.237 \rightarrow -0.020$. Diagnosis: training loss plateaus at ~ 1.06 after ~ 900 steps, gradient median stays ≈ 0.25 for 4,700 further steps with early spikes (max 547), and fold classification never leaves chance (train loss $\approx \ln 7$ throughout). This is joint-optimization instability landing in bad basins, not overfitting: the training loss never approaches zero.

Stabilization repairs, and confirms a plateau. Warmup 0.05→0.15, clip 1.0→0.5, and dev early stopping (every 200 steps, best-dev promotion) recover the 5,600-update budget to 0.481 — statistically the small budget’s 0.485 (gain vs init +0.049 $\approx$ +0.050). Beyond $\approx$ 2,500 updates, compute is not the bottleneck; the ceiling is capacity/interference of joint multi-task training (Figure 3a).

Per-family early stopping is free. On one training trajectory, family-optimal dev peaks span steps 200 (GUE) to 5,200 (local snapshots) — a $26\times$ spread — while the global best sits at 2,200. Selecting each family’s own best-dev checkpoint dominates the single global checkpoint at family level (all $\Delta \geq 0$; mean +0.063; per task 21/9/7 wins/ties/losses; Figure 3b,c), at zero inference-time cost. Part of the plateau was a checkpoint-selection compromise, not pure capacity.

5.4 Controlled mechanism ablations

Beyond the phrasing ablation of Section 5.2d (candidate surface form is a real but run-dependent degree of freedom, and no phrasing closes the noul gap), one more probe concerns frozen candidate-likelihood scoring: on a multimodal route-A pilot (synthetic property maps, frozen Qwen2.5-VL (Bai et al., 2025)), plain candidate likelihood collapses to one class (accuracy 0.040 < chance), while *contrastive decoding* $\log p(\text{cand} \mid \text{img}, q) - \log p(\text{cand} \mid q)$ (cf. Leng et al., 2024; Li et al., 2023) restores multi-class predictions but stays at chance (Figure 4b). The reusable lesson: frozen-VLM candidate-likelihood evaluation should default to contrastive decoding; the route itself is rejected (Appendix C).

5.5 Full-benchmark family-specialized baselines

The joint-37 comparison of Section 5.1 is a deliberate stress test; it is not what the benchmark can support under a practical recipe. To calibrate the latter — and to give the community a starting leaderboard — we train each family *separately* on the biology-CPT encoder init (Section 4.1 footnote: the CPT arm) with the same stabilized protocol and per-family dev early stopping, for both decision architectures, and evaluate every supported task’s test split (Table 10, single training seed; the variance caveats of Section 4.3 apply and are why the matched comparison above is seed-averaged).

Three findings. *First, specialization plus CPT recovers near-SOTA numbers:* TAPE fluorescence reaches Spearman 0.627 with task heads (0.568 with the shared scorer), against the published ceiling of 0.68; dna_core reaches accuracy 0.625 (from 0.475 under joint no-CPT training); and the local-snapshot family reaches 0.860 on nuclear-protein positioning, 0.855 on signal-peptide detection and 0.785 on promoter detection (single-task specialized recipes reach 0.88–0.91 on the latter); and Genomic-Benchmarks proposition tasks reach AUROC 0.897/0.839/0.834 on coding-vs-intergenic, human-or-worm and non-TATA promoters. Not every family improves: GUE’s 28 tasks average 0.497 accuracy under specialization (best 0.570), and the virus_covid task scores 0.115 — its test split carries 1,754 quarantined conflicting-label groups and an adversarial class balance, which the leaderboard reports rather than hides. ProteinGym is the starkest headroom: 185 of 217 assays are evaluable within the 512-token budget, and family-specialized training reaches only 0.012 macro Spearman with task heads (−0.022 with the shared scorer; median 0.017, best 0.399) against MSA-augmented supervised ceilings near 0.55 — single-sequence mutation-fitness prediction at this sample and update budget remains open, and the benchmark quantifies the gap. The low absolute scores of Section 5.1 are therefore a property of the joint small-budget regime and the encoder init, not of the tasks being unlearnable. *Second, the architecture ordering replicates on the score interface:* on TAPE stability, per-task heads reach 0.433 while the shared scorer’s five-anchor distribution reaches 0.049 — the same heads-over-shared-scorer ordering as the seed-averaged noul result, now on a different primitive and a different training regime, which argues against the main comparison being an artifact of joint training. *Third, the benchmark is not saturated:* DeepSTARR (0.23/0.21 vs. published PCC 0.68/0.74; our 512-token window truncates its $\sim$ 2kb inputs), the legacy splice/TF blind tests (constant-prediction collapse in both arms), and several ProteinGym assays remain far below published bests (Table 11), so the benchmark discriminates and leaves measurable headroom.

The cost of unification. One checkpoint trained on all 233 gated tasks (Table 10, “single” columns) serves classification and proposition families nearly losslessly (Genomic-Benchmarks noul AUROC 0.698 vs 0.715 specialized) but sacrifices small regression families (TAPE 0.299 vs 0.530); the heads-over-shared-scorer ordering holds here as well (0.698 vs 0.513), the fourth regime in which the main comparison replicates.

Granularity gradient. Five tasks fine-tuned individually (Figure 5) reach 0.845 promoter, 0.671 fluorescence (99% of the published ceiling) and 0.892 coding AUROC; per-task ordering across the four granularities shows within-family positive transfer (family-joint $\geq$ single for npp/fold/coding) and cross-family interference (single $>$ all-241 $>$ 37-slice for promoter/fluorescence) — the evidence base for family-balanced scheduling and per-family checkpoints (Section 4.2). Per-family, per-task and prior-best numbers: Tables 10 and 11 and the released CSV.

6 Discussion and Limitations

When would a shared typed-decision scorer win? Within our evidence: not on seen tasks, not zero-shot (at base level), weakly on cross-family classification only, and its zero-adaptation edge is erased by ≈ 32 labels. What survives unambiguously is the interface itself: no task parameters, arbitrary dynamic candidates, immediate inference on new tasks (#2 architecturally cannot), and inference-time degrees of freedom (phrasing, per-family checkpoints) — though the 3-seed ablation shows phrasing tuning does not close the noul gap ($0.553 < 0.629$). Deployment regimes with very large or open candidate sets, frequent task arrivals, or true zero-label cold starts may shift the balance; those are hypotheses, not our results.

Absolute levels vs. specialized anchors. Joint-37 choice ≈ 0.49 and score ≈ 0.01 – 0.02 sit far below specialized training: the same architecture family reaches 87–89% on promoters, 56–58% on fold class and GFP MAE 0.32–0.49 under per-task recipes (Wang, 2024). The joint regime is a deliberate stress test (one checkpoint serving 37 heterogeneous tasks, ≤ 2048 samples each, ≈ 70 updates per task); it calibrates “where shared decision interfaces currently stand under multi-task small budgets” and is not an architecture ceiling.

Threats to validity. (i) The head-to-head uses a 37-task slice (PG/GUE capped at 12), not all 290: the benchmark is the infrastructure contribution and the comparison is a matched slice — we state the two scopes separately everywhere. (ii) 3 seeds; SEM is across tasks, run-std across seeds; noul ($n=7$) and score ($n=9$) have few tasks. (iii) #3 is not matched (frozen, different backbone, truncation, different score protocol). (iv) The cross-family score arm now has 3 seeds, but with $n=2$ tasks the stability $\Delta = +0.159$ depends on the pathological frozen baseline and must be read with that caveat (Section 5.2). (v) dev selection noise is mitigated (96 $\rightarrow$ 256 samples) but not eliminated. (vi) GPU nondeterminism is irreducible; only seed averaging absorbs it. (vii) the 512-token budget excludes long-sequence examples (exclusion rates reported, never silent truncation).

7 Conclusion

We introduced BioDecisionBench — 290 biological sequence decision tasks from 10 source families, three modalities and four typed decision primitives, under one record format and a leak-proof admission protocol (zero within-task leakage after isolation; conflict groups quarantined, legacy blind tests preserved verbatim) — together with reproducible build/evaluation/training code. On its 37-task matched slice, the first seed-averaged three-way architecture comparison gives a bounded, honest answer: the shared typed-decision scorer does not beat per-task heads (significantly worse noul, parity elsewhere, less stable), its zero-shot and cross-family transfer are weak, and its zero-adaptation claim is matched by a 32-label fresh head. Its real value is the interface: dynamic candidates, no task parameters, immediate new-task inference. Methodologically, we documented a complete conclusion-reversal case from run variance (± 0.05 – 0.1) and propose seed averaging, run-std reporting and stabilized selection as the minimum bar for architecture comparisons in this regime. Future work: per-family training over all 290 tasks, per-task early stopping, a matched generative arm (fine-tuned small LM), real-image multimodal evaluation, and re-testing transfer at larger budgets and backbones.

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Table 5: Per-family, per-seed means for the two trained arms (task-macro of per-task seed means). #1 uses per-family early-stop checkpoints (37-task slice; families as in the training scheduler), #2 the full-run checkpoint. DeepLoc (multi_noul) and double-sequence tasks are outside this harness.

Family	Primitive	#1 (mean $\pm$ std over 3 seeds)	#2	n
DeepSTARR	score	0.035 $\pm$ 0.027	0.053 $\pm$ 0.023	3
GUE	choice	0.464 $\pm$ 0.017	0.478 $\pm$ 0.013	3
GenomicBenchmarks	choice	0.402 $\pm$ 0.082	0.305 $\pm$ 0.014	3
GenomicBenchmarks	noul	0.478 $\pm$ 0.017	0.629 $\pm$ 0.005	3
ProteinGym	score	0.015 $\pm$ 0.058	0.022 $\pm$ 0.015	3
TAPE	score	-0.014 $\pm$ 0.047	-0.027 $\pm$ 0.043	3
dnagpt_pools	choice	0.519 $\pm$ 0.006	0.514 $\pm$ 0.030	3
local_snapshots	choice	0.534 $\pm$ 0.016	0.536 $\pm$ 0.030	3

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A Full per-family results

Table 5 reports per-family, per-seed means for the two trained arms.

B Adaptation cost and ablation tables

C Multimodal route-A pilot

Protein sequences without resolved structures were rendered as synthetic amino-acid property heatmaps (7 normalized physicochemical tracks $\times$ position, matplotlib viridis) and presented to a frozen Qwen2.5-VL-3B (Bai et al., 2025) alongside the fold-class question. Plain full-candidate likelihood collapsed to a single candidate on all 200 images (accuracy 0.040; macro-F1 0.011; 1/7 classes predicted) — candidate text priors dominate and the image does not change the ranking. Contrastive decoding, $\log p(\text{cand} \mid \text{img}, q) - \log p(\text{cand} \mid q)$, restores multi-class behavior (4/7 classes, accuracy 0.140) but sits at chance ($1/7 = 0.143$), far below the majority baseline (0.325) and the text-only frozen LM (0.320). Image sanity checks rule out rendering failure (per-image statistics and cross-image correlations normal). Conclusion: route A (synthetic renders) with a frozen VLM is a dead end;

Table 6: Adaptation cost on 11 held-out tasks: #1 zero-shot (no labels, no training) vs #2 new linear head fitted on N labels (frozen shared encoder). #2 catches or beats #1’s zero-shot on 7/11 tasks already at $N=32$.

Task	Metric	#1 zero-shot	#2 $N=32$	$N=128$	$N=512$
gue_H3	accuracy	0.555	0.510	0.510	–
gue_H3K14ac	accuracy	0.445	0.543	0.543	–
gue_H3K4me1	accuracy	0.430	0.532	0.532	–
gue_H3K4me2	accuracy	0.450	0.541	0.541	–
gue_H3K79me3	accuracy	0.490	0.492	0.492	–
gue_H3K9ac	accuracy	0.485	0.451	0.451	–
pg_ACE2_HUMAN_Chan_2020	spearman	-0.176	0.029	-0.005	0.040
pg_B2L11_HUMAN_Dutta_2010_binding-Mcl-1	spearman	0.133	-0.054	–	–
pg_CATR_CHLRE_Tsuboyama_2023_2AMI	spearman	0.066	-0.086	-0.039	0.135
pg_CSN4_MOUSE_Tsuboyama_2023_1UFM	spearman	-0.034	0.171	0.101	-0.088
pg_F7YBW8_MESOW_Ding_2023	spearman	-0.112	0.416	–	–

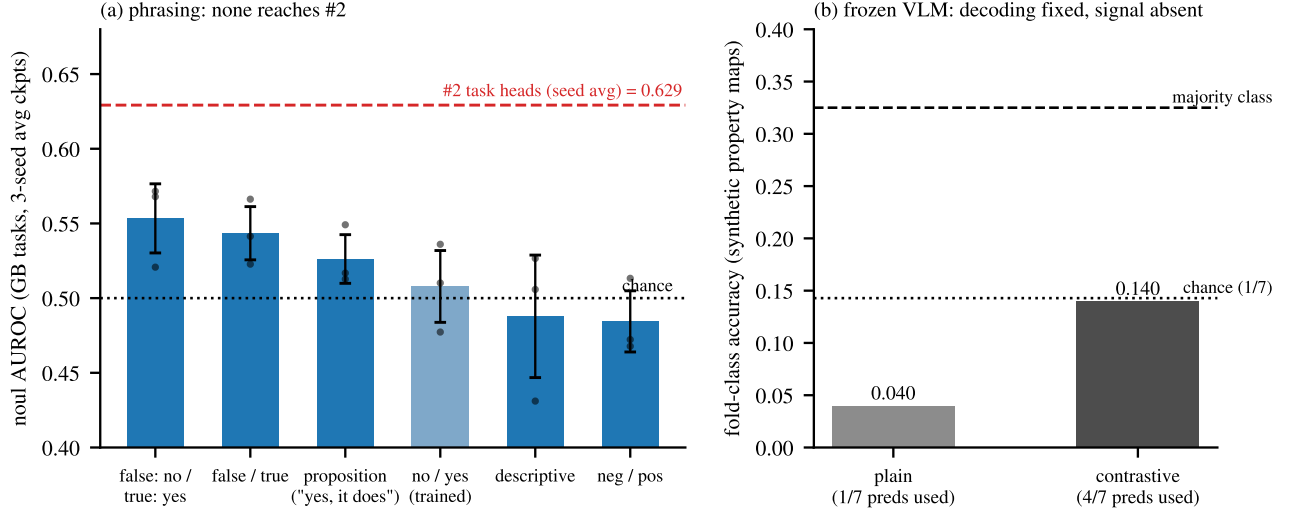


Figure 4: Controlled mechanism ablations. (a) Noul candidate phrasing on three seed-averaged stable checkpoints (same weights; only candidate text varies). No phrasing reaches the per-task-head reference (0.629, dashed); the best (“false: no / true: yes”, 0.553 ± 0.023) beats the *trained* phrasing (hatched) on all three seeds — reversing the single-checkpoint finding, i.e., inference-level ablations are also checkpoint-dependent. (b) Frozen VLM on synthetic property maps (multimodal route-A pilot): plain candidate likelihood collapses to one class (acc 0.040, 1/7 classes used); contrastive decoding removes the collapse (0.140, 4/7) but remains at chance — decoding fixed, signal absent.

the reusable finding is that *frozen-VLM candidate-likelihood evaluation should default to contrastive decoding*. Real-image localization (HPA) and fine-tuned VLM arms are a separate work.

D CPT-arm main comparison

E Granularity gradient

F Leaderboard tables

G Data, licensing, and disclosures

Data and licensing. All sources are public datasets with license, revision and SHA-256 recorded in download receipts; no human-subject data. RNAcompete is the 14-RBP test subset (not the full ~ 200); DeepGOZero/PEER/DeepSEA remain in the catalog as needs-upstream-work and are not claimed as coverage. The splice label-convention doubt (Section 3.3) is explicitly recorded for independent verification.

Table 7: Noul AUROC on GB tasks for six candidate phrasings, evaluated on three seed-averaged stable checkpoints (same weights per column; only rendered candidate text varies). No phrasing reaches the per-task-head reference (0.629). The trained phrasing is marked †.

Phrasing	s1	s2	s3	mean±std
false: no / true: yes	0.572	0.568	0.521	0.553±0.023
false / true	0.566	0.523	0.541	0.543±0.018
proposition (“yes, it does”)	0.517	0.513	0.549	0.526±0.016
no / yes (trained)†	0.477	0.536	0.510	0.508±0.024
descriptive	0.527	0.506	0.431	0.488±0.041
neg / pos words	0.513	0.468	0.472	0.484±0.020

Table 8: Cross-family score transfer (TAPE held out; DeepSTARR and ProteinGym score families in training). Spearman per seed; “abs” = trained absolute value. Stability’s positive Δ reflects a pathological frozen baseline (−0.218) regressing toward zero, not genuine transfer.

Task		frozen	tr. s1	tr. s2	tr. s3	Δ mean±std
tape_fluorescence	abs	0.037	-0.008	0.026	-0.075	-0.056±0.042
tape_stability	abs	-0.218	-0.018	0.020	-0.179	+0.159±0.086

LLM-use disclosure. [To be finalized per venue checklist: writing assistance disclosure.]

H Reproducibility

Every figure and table regenerates from on-disk JSONs via one script per artifact (`paper_v2/figures/gen_*.py`); no metric is hardcoded in the plotting or table code. Each run persists `run_config.json` (flags, seeds, caps), `train_trace.jsonl` (per-step loss/grad) and `dev_curve.jsonl` (per-family dev scores); evaluation subsets are drawn with a fixed sampling seed (20261001) rather than first- N — a bug we found and fixed: class-ordered test files plus first- N sampling produced a spurious noul AUROC of 0.387 that looked like a polarity bug but was a sampling artifact. Hardware: single H20/A800-class GPU per run; nondeterminism is absorbed by the seed protocol of Section 4.3.

Table 9: Seed-averaged three-way comparison on the biology-CPT encoder init (3 seeds, same protocol as Table 4). The architecture ordering is invariant to the encoder init: per-task heads match or beat the shared scorer on every primitive under both inits; CPT lifts both arms on noul (+0.041/+0.022) without closing the gap.

Primitive	Metric	n	#1 shared (cpt)	#2 heads (cpt)	#3 frozen
choice	acc	21	0.485 $\pm$ 0.030	0.500 $\pm$ 0.031	0.424
noul	auroc	7	0.519 $\pm$ 0.020	0.651 $\pm$ 0.042	0.522
score	spear	9	-0.001 $\pm$ 0.011	0.056 $\pm$ 0.037	–
Per-task seed-mean wins (#1 / tie / #2)					3 / 22 / 12

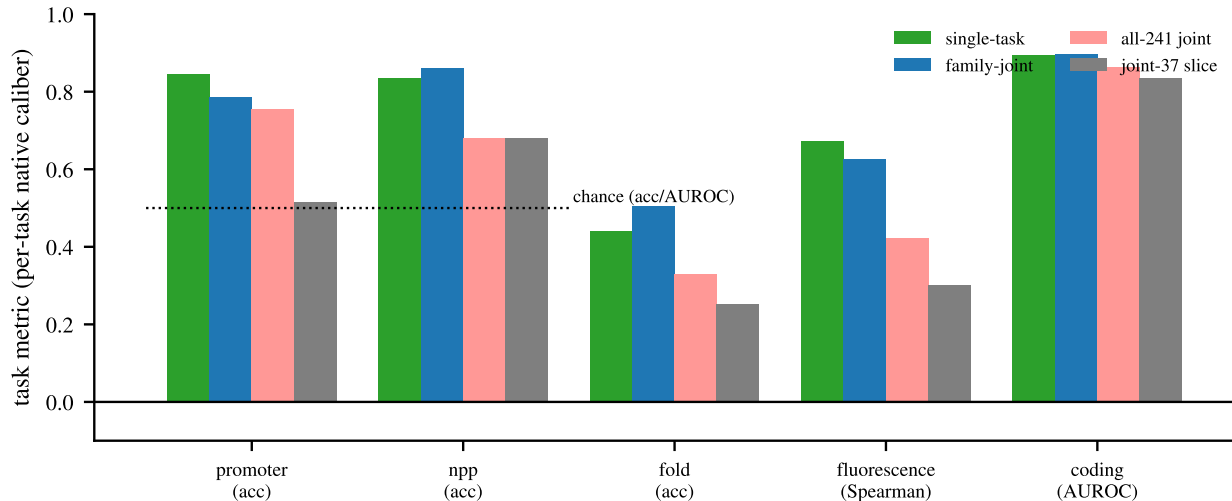


Figure 5: Granularity gradient (#2 arm, CPT init, single seed): five representative tasks under four training granularities. Within-family joint training matches or beats single-task fine-tuning (positive transfer), while cross-family joint training degrades monotonically with scope (interference) — except ProteinGym-scale score tasks, which stay near zero at every granularity.

Table 10: Three training regimes over the full supported benchmark (single training seed; stabilized protocol; global best-dev early stop for the single-model columns, per-family early stop for the specialized columns): one unified model over all ~ 243 gated tasks (single), vs one model per family (spec.). Task-macro of accuracy (choice), AUROC (noul), Spearman (score). The single-vs-spec. gap is the measured cost of unification; compare also the joint-37 slice in Table 4.

Family	Primitive	Tasks	#1 single	#2 single	#1 spec.	#2 spec.
dnagpt-pools	acc	3	0.402	0.562	0.527	0.547
TAPE	Spearman	2	0.067	0.299	0.309	0.530
DeepSTARR	Spearman	2	0.072	0.176	0.077	0.217
local-snapshots	acc	5	0.516	0.603	0.525	0.684
GenomicBenchmarks	acc	1	0.390	0.485	0.575	0.515
GenomicBenchmarks	AUROC	7	0.513	0.698	0.641	0.715
GUE	acc	28	0.485	0.494	0.492	0.497
ProteinGym	Spearman	193	-0.007	0.008	-0.022	0.012

Table 11: Published bests vs our family-specialized CPT baselines (#2, single seed). Metric calibers differ per family and are *never* averaged across rows; the note column states comparability. Sources transcribed from the cited papers’ tables (see `prior_best.json`); unverified numbers are excluded.

Family	Ours	Literature best (their metric)	Note
GUE	0.497 (acc)	DNABERT-2 67.77 / NT-2.5B 66.93	ours is accuracy; not directly comparable, caliber ref only
GenomicBenchmarks	0.515, 0.715 (acc/AUROC)	HyenaDNA best on 7/8 (e.g. worm 96.6, regu- latory 93.8)	comparable (acc)
ProteinGym	0.012 (Sp)	ProteinNPT 0.547 (supervised, fold-contiguous split)	comparable: same official split family
TAPE	0.530 (Sp)	fluor 0.68 / stab 0.73 (TAPE- era pretrain; ProtBERT-ft 0.678/0.734)	comparable
DeepSTARR	0.217 (Sp)	0.68 / 0.74 (au- thors’ CNN)	PCC vs our Spearman; 512-token win- dow truncates 2kb inputs
dnagpt-pools	0.547 (acc)	LLaMA-Gene category-level 0.83 (DNA class.)	category-level only; per-dataset numbers unverifiable locally
local-snapshots	0.684 (acc)	self-prior: pro- moter 0.911, fold 0.600 (working ms Table 2)	same-lineage prior; internal record, not peer-reviewed